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Dealing with high uncertainty in qualitative network models using Boolean analysis

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Abstract

1. Models for predicting ecological behaviour typically require large volumes of data for parameterisation, which is a problem because data are scarce. Qualitative modelling (QM) provides an alternative by exploring the entire range of possible parameter values. When a parameter value is completely unknown, QM typically invokes the Principle of Indifference (PoI), for example by sampling the parameter from a uniform prior distribution. However, if PoI is invoked in this probabilistic way, there may be multiple possible methods for defining a parameter space and sampling values from it, and, worryingly, two different but equally defensible methods can lead to different predictions about ecosystem responses.
2. We investigated how probabilistic PoI can give rise to problems in QM, and developed an alternative method based on Boolean PoI that does not suffer the same limitations. We used a case study that involved predicting the responses of multiple species to the suppression of a pest. The unknown model parameters were interaction strengths between species. For the standard probabilistic method, we drew the parameters randomly from uniform (PoI) and other distributions. For our new Boolean PoI method, we instead simply specified the ranges of "possible" parameter values, and developed a Boolean analysis technique to summarise model predictions.
3. As expected, invoking probabilistic QM yielded different predictions (species-response probabilities) for different but equally defensible parameterisation and sampling schemes. Sometimes differences were large enough to impact decision-making. In contrast, our new Boolean PoI approach simply classifies outcomes (species responses) as certain, possible, or impossible. Encouragingly, some species responses that were not consistently resolved under probabilistic PoI were shown by our method to be in fact governed by simple rules. Our method can also identify key species whose responses determine whole-system outcomes.
4. Our non-probabilistic representation of uncertainty circumvents the philosophical problems in standard implementations of PoI for QM. Our Boolean analysis method summarises results in a way that is interpretable and potentially useful to conservation decision makers. A priority for future research is to increase the efficiency of our Boolean approach to allow it to deal with problems of higher complexity (more interactions).

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KEYWORDS
belief probabilism, Boolean minimisation, community matrix, ensemble ecosystem modelling, loop analysis, press perturbation, sensitivity matrix, weighted-predictions matrix

1 | INTRODUCTION

Conservation decision makers face the problem that the management options available to them may lead to unexpected negative outcomes. For example, pest control can trigger mesopredator release (Courchamp, Langlais, & Sugihara, 1999), release of a competitor (Ruscoe et al., 2011) or more complex feedbacks between species, leading to even worse outcomes for native species. Predictions of these outcomes are needed for sound decisions to be made.

Complex outcomes can in principle be predicted with dynamical systems models (e.g. Courchamp et al., 1999; Levins, 1974), but in most applications such models are impossible to parameterise due to a lack of data. Data are often lacking about which species and interactions are present, that is structural uncertainty, and the strengths of interactions between species, that is interaction—strength uncertainty. Both forms of uncertainty can have profound effects on model predictions (e.g. Marzloff, Dambacher, Johnson, Little, & Frusher, 2011; Novak et al., 2011). Structural uncertainty is typically resolved by expert elicitation, but interaction—strength uncertainty is more problematic. Interaction strengths are often completely unknown, and to quantify them empirically would require perturbing every species' population separately while monitoring every other species (e.g. Schmitz, 1997). This is rarely attempted because of the difficulty involved (e.g. Dambacher, Li, & Rossignol, 2003, reports only two examples).

Recently, modellers have turned to a suite techniques, which we refer to collectively as qualitative modelling (QM), that hold the promise of obtaining predictions from dynamical models even when the data required to parameterise them are lacking (Melbourne-Thomas, Wotherspoon, Raymond, & Constable, 2012; Rochet & Rice, 2009; Rochette, Lobry, Lepage, & Boët, 2009). QM achieves this by exploring the range of parameter value uncertainty to obtain a range of predictions, then interpreting the predictions qualitatively or probabilistically. For example, if 89% of model simulations in the uncertainty range predict that pest control will increase petrel population size, then that is interpreted as high support for that prediction (Raymond, McInnes, Dambacher, Way, & Bergstrom, 2011).

Early QM approaches to interaction—strength uncertainty produced deterministic predictions, but they were later extended to produce probabilistic predictions as well. Loop analysis (Lane & Levins, 1977; Levins, 1974; Marzloff et al., 2011) assumes that population dynamics are near equilibrium, and that a press-perturbation can be used to simulate the scenario of interest (e.g. pest control) (Bender, Case, & Gilpin, 1984; Nakajima, 1992; Yodzis, 1988). Loop analysis can identify species responses that will occur regardless of what the interaction-strength values are, classifying responses as positive, negative or indeterminate if either response is possible. The weighted-predictions matrix approach builds on loop analysis, by inferring probabilities for the indeterminate species responses (Dambacher et al., 2003; Lassalle et al., 2013; Metcalf, Pember, & Bellchambers, 2011; Ramos-Jiliberto, Garay-Narváez, & Medina, 2012; Wildermuth, Fay, & Gaichas, 2018). Weightings in the matrix count the proportion of positive versus negative feedback loops between species, and high weightings are interpreted as evidential support for an indeterminate response taking a particular sign. More recently, Monte Carlo simulations have been used (Harvey, Reum, Poe, Williams, & Kim, 2016; Raymond et al., 2011; Reum et al., 2015; Sobocinski, Greene, & Schmidt, 2018), where the proportions of random parameterisations giving a positive or negative species response are interpreted probabilistically. The probabilistic approach can also address structural uncertainty, by comparing the likelihoods with which candidate network structures reproduce past observations (Hansen, Tunney, Winslow, & Vander Zanden, 2017; Hosack, Hayes, & Dambacher, 2008; Melbourne-Thomas et al., 2012; Montaño-Moctezuma, Li, & Rossignol, 2007). Conceptually related approaches may include fuzzy cognitive maps (e.g. Baker, Holden, Plein, McCarthy, & Possingham, 2018), ensemble ecosystem modelling (e.g Baker, Gordon, & Bode, 2017), and Monte Carlo simulations in other types of models (e.g. Lassalle et al., 2013). To varying degrees, these methods express a philosophical idea: ignorance can be expressed in probabilistic terms, and the outcomes of doing so, also expressed as probabilities, quantify one's confidence in a prediction.

When there is complete ignorance of parameter values, QM implicitly invokes the Principle of Indifference (PoI) (Keynes, 1921). PoI encompasses two ideas (Norton, 2008). The first idea (PoI 1) is a truism of evidence: if we have no grounds for distinguishing possible scenarios, then we should favour them equally. The choice to favour scenarios equally can be justified on epistemic grounds, expressing our degree of belief, or objective grounds, expressing the degree of support from evidence. PoI 1 is invoked, for example, when equal weighting is given to each positive and negative feedback loop in the weighted-predictions matrix. The second idea (PoI 2) is that the extent of favouring can be represented by a probability. PoI 2 is invoked, for example, by sampling unknown interaction strengths from a uniform prior, and by interpreting predictions with higher probabilities as having higher support.

The two ideas of PoI work well in canonical examples (e.g. tossing a die); however, we encounter philosophical and technical difficulties when background information is lacking (Box 1). The difficulty arises because we must describe a parameter space to represent the equally favoured scenarios, and there are usually multiple ways to do so. In cases of very little background knowledge, we may have no means to privilege one description of the parameter space as the correct one. This can lead to a paradoxical result: different but equally defensible descriptions predict different probabilities for the same outcome (e.g. Bertrand's paradox (reviewed in Shackel, 2007); the squares factory (Smith, 2014)). This general problem with PoI

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BOX 1. Principle of Indifference examples

Example 1: Tossing a die

A die is tossed. We have no reason to favour any one outcome over any other, therefore by PoI 1 we must treat each outcome 1, …, 6 equally. By PoI 2, we assign the same probability, $\frac{1}{6}$, to each. This gives the correct (observed) probabilities.

Example 2: The von Mises wine/water paradox (reviewed in Deakin, 2006)

All that is known about a bottle of liquid is that it contains a mix of wine and water, and that the ratio of wine to water is between 1/3 and 3. What is the probability that the ratio of wine to water is less than or equal to 2? Assigning the same probability to each value is equivalent to sampling from a uniform distribution. If we define $x = \frac{\text{wine}}{\text{water}}$, then $P(x \leq 2) = (2 - 1/3)/(3 - 1/3) = 5/8$. However, we can reframe the problem in terms of $y = \frac{\text{wine}}{\text{water}}$. Then, applying PoI gives $P(y \geq 1/2) = (3 - 1/2)/(3 - 1/3) = 15/16$. These are equivalent and equally defensible descriptions of the parameter space, but applying PoI to each of them gives contradictory probabilities.

Example 3: Modelling for conservation decision-making

An expert elicitation workshop is being conducted, to develop a model (similar to Figure 1) to predict the effects of pest suppression on other species in an ecosystem. Experts have produced an interaction network structure, and conservation decision makers have agreed to use a population dynamic model and its near-equilibrium assumptions. However, the experts state that no data exist to quantify the interactions between the species. They have no reason to prefer any values over any others; therefore, the modellers propose that they will treat all possible values equally.

The modellers consider two approaches: (a) the community matrix approach, where a uniform prior is used to sample interaction-strength values equally; or (b) an approach similar to the weighted-predictions matrix, where each positive and negative feedback loop is weighted equally. Which should they use?

The modellers could themselves choose an approach. Both are equally valid representations of the same model and ways to represent the interactions. However, the modellers know that the different approaches will produce different prediction probabilities, which can be different enough to impact conservation decision-making. The modellers could clarify that the choice is an additional assumption underlying the model predictions. However, the conservation decision makers find the assumption's meaning obscure: if both are equally valid, then what does it mean to choose one or the other, and how do they account for that in their decision-making?

The modellers could instead ask the experts to choose. However, when the experts said that they had no data about interaction-strength values that included not knowing which parameter space would be best to represent that lack of data. Furthermore, the conservation decision makers say that this ambiguity is part of the uncertainty that they need the modelling approach to account for, in order to make their decision.

The modellers decide to present predictions from both approaches separately— and this seems to address the issues above. They have captured experts' uncertainty about both the parameter value and parameter space. Where the predictions agree, the decision makers feel more confident that they understand the risks of their management interventions; and where they disagree, it suggests that the combined approach is doing a good job of capturing all of the uncertainty present.

However, the exercise has given the modellers pause. They know that there are many additional model representations, also having mathematical equivalence to the representations above, potentially including ones that they are not aware of. Each has a different parameter space and can therefore produce different predictions. Furthermore, it is paradoxical that two equally valid ways of representing the same thing—'ignorance'—should produce different results (c.f. wine/water paradox); it suggests that information is missing from the problem specification. In simple examples (e.g. die roll), a parameter space can be chosen using background information about the mechanism of randomisation, however the mechanisms determining species interaction strengths are too complex to resolve in this way.

raises the possibility that QM that implicitly invokes PoI may suffer from similar ambiguity problems, and that this may impact conservation decision-making.

In this paper, we will use a case study involving interaction—strength uncertainty (Raymond et al., 2011) to illustrate how different but equally defensible random parameterisation methods lead to contradictory predictions. We will then turn to the philosophy literature for a non-probabilistic method of expressing uncertainty under PoI, which circumvents the philosophical problems described above. We will demonstrate how Boolean analysis can be used to summarise non-probabilistic model predictions into a form that is useful to conservation decision makers.

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2 | BACKGROUND

2.1 | Background to the Macquarie Island case study

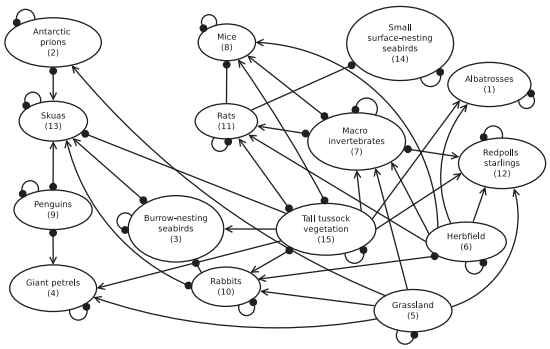
Raymond et al. (2011) used a population dynamic model to predict the responses of species on Macquarie Island, Australia, to pest control of rabbits, rats and mice. They simulated pest control as a negative press perturbation using the sensitivity matrix approach (Bender et al., 1984; Nakajima, 1992; Yodzis, 1988). The sensitivity matrix predicts the change in the population sizes of other species, and the changes can be summarised by their signs (Figure 1).

In order to obtain predictions of species responses in this model, prior knowledge of the interaction strengths between all species is needed, but because these were not known, Raymond et al. (2011) used Monte Carlo simulations. For each network structure (obtained from expert elicitation, e.g. Figure 1a), they sampled interaction-strength magnitudes from a uniform distribution and created an ensemble of community matrices (e.g. Figure 1b). This ensemble was filtered using plausibility constraints: the population dynamics must be locally stable, and the model must reproduce past observations (e.g. suppression of rabbits increases tall tussock vegetation). For each plausible community matrix, they derived a sensitivity matrix (e.g. Figure 1c) predicting species responses to pest control (e.g. Figure 1d). Then, for each species, they interpreted the proportion of responses of a particular sign as a probability indicating the level of support for that response occurring in the real system.

2.2 | Background to the Boolean approach

The problems of PoI cannot be resolved by discarding PoI 1 (equal favouring of indistinguishable scenarios) because it is a truism; therefore, the solution must be found by relinquishing PoI 2 and using a non-probabilistic representation of favouring instead (Norton, 2008). When the full continuous-valued probability framework is

(a) interaction network



(b) community matrix A

$$\begin{bmatrix} -a_{1,1} & 0 & 0 & 0 & 0 & \cdots & a_{1,15} \\ 0 & -a_{2,2} & 0 & 0 & a_{2,5} & & 0 \\ 0 & 0 & -a_{3,3} & 0 & 0 & & a_{3,15} \\ 0 & 0 & 0 & -a_{4,4} & a_{4,5} & & a_{4,15} \\ 0 & 0 & 0 & 0 & -a_{5,5} & & 0 \\ \vdots & & & & & \ddots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \cdots & -a_{15,15} \end{bmatrix}$$

(c) sensitivity matrix $S = (-A)^{-1}$

$$\begin{bmatrix} s_{1,1} & s_{1,2} & s_{1,3} & s_{1,4} & s_{1,5} & \cdots & s_{1,15} \\ s_{2,1} & s_{2,2} & s_{2,3} & s_{2,4} & s_{2,5} & & s_{2,15} \\ s_{3,1} & s_{3,2} & s_{3,3} & s_{3,4} & s_{3,5} & & s_{3,15} \\ s_{4,1} & s_{4,2} & s_{4,3} & s_{4,4} & s_{4,5} & & s_{4,15} \\ s_{5,1} & s_{5,2} & s_{5,3} & s_{5,4} & s_{5,5} & & s_{5,15} \\ \vdots & & & & & \ddots & \vdots \\ s_{15,1} & s_{15,2} & s_{15,3} & s_{15,4} & s_{15,5} & \cdots & s_{15,15} \end{bmatrix}$$

(d) species response

- If $\mathrm{sgn}(s_{1,10}) = +$, then albatrosses predicted to respond negatively to rabbit control;
- If $\mathrm{sgn}(s_{1,10}) = -$, then albatrosses predicted to respond positively to rabbit control;
- If $\mathrm{sgn}(s_{2,10}) = +$, then antarctic prions predicted to respond negatively to rabbit control;
- If $\mathrm{sgn}(s_{2,10}) = -$, then antarctic prions predicted to respond positively to rabbit control;
- ...

FIGURE 1 General procedure for modelling the effect of a pest control programme on species in an ecosystem. The interaction network describes the direct interactions between species, where positive effects are indicated by an arrow and negative effects by a filled circle (a). In the Macquarie Island case study, interaction networks were obtained using expert elicitation. The corresponding community matrix has elements that quantify the strengths of the interactions between species (b). In the case study, the interaction strengths were unknown, so numerical values were randomly sampled. Pest control is modelled as a negative press perturbation on the pest species; therefore, the sensitivity matrix, which is calculated as the inverse of the community matrix, determines the response of species to pest control (c). Species responses to pest control are summarised by their sign that is 'positive' or 'negative' (d)

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replaced with a three-valued system, such that scenarios are classified as simply (1) possible, (2) necessary, or (3) impossible, then the PoI paradoxes no longer occur (Norton, 2010a,b). The next challenge is to turn this impoverished description into useful outputs for decision makers.

When the case study's model inputs—the interaction strengths—are unknown, our non-probabilistic approach will treat every input value as possible; however, this does not mean that every output—every species—response combination (e.g. albatrosses respond positively, prions respond negatively, etc.)—is possible. Some combinations will be impossible, regardless of what the interaction-strength values are, because of the structure of the interaction network. The classic example is a food chain: a negative press-perturbation of the apex predator always has a positive effect on its prey, which in turn has a negative effect on that prey's prey, and so on with alternating signs down the chain (Usmani, 1994); therefore, it is impossible for a predator–prey pair to have the same response sign. More complex interaction networks will have more complex rules, but the principle remains the same: the network structure determines the ecological feedbacks between species, which makes certain species—response combinations impossible. Plausibility constraints, for example representing past observations, will further increase the set of impossible combinations.

Our approach begins with a truth table (e.g. Figure 2a), which lists every species—response combination that is combinatorially possible. We then classify each as possible in the model if it can arise under at least one parameterisation, or impossible otherwise. As the number of species responses grows, the truth table quickly becomes very long (*n* responses gives 2^n rows) and difficult to interpret. A method for summarising the truth table is needed.

The truth table can be summarised using Boolean analysis, which is a technique for analysing logical relationships between dichotomous variables (Theuns, 1994). It is used in sociology for analysis of questionnaire data, to determine if a 'yes' or 'no' answer to a question is always predicted by some combination of 'yes' or 'no' answers to another set of questions (e.g. Degenne & Lebeaux, 1996). In the context of pest control, this is analogous to determining if a positive or negative response in a species is always predicted by some combination of positive or negative responses in another set of species.

The first step of Boolean analysis is Boolean minimisation, which reduces the truth table to the minimal table needed to provide its

(a) truth table

Negative perturbation of spp 3 causes positive response in: species 3	species 4	species 5	Species-response combination impossible in the model?
False	False	False	True
False	False	True	True
False	True	False	True
False	True	True	False
True	False	False	False
True	False	True	True
True	True	False	True
True	True	True	True

(b) minimised truth table

Negative perturbation of spp 3 causes positive response in: species 3	species 4	species 5	Species-response combination impossible in the model?	
False	-	False	True	i
-	False	True	True	j
True	True	-	True	k

(c) logical relationships between species-responses

i: if spp 3 responds negatively then spp 5 responds positively
if spp 5 responds negatively then spp 3 responds positively

j: if spp 4 responds negatively then spp 5 responds negatively
if spp 5 responds positively then spp 4 responds positively

k: if spp 3 responds positively then spp 4 responds negatively
if spp 4 responds positively then spp 3 responds negatively

(d) logical implication network

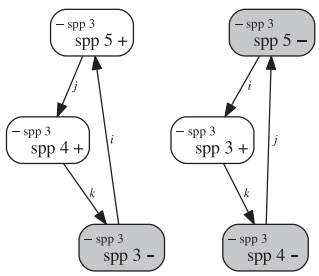

FIGURE 2 An example of Boolean analysis, using the results from model simulations of three species' responses to a negative press perturbation of 'species 3' (details in Supporting Information A). Every species–response combination that is combinatorially possible was classified as either possible or impossible in the model and encoded in a truth table (a). Boolean minimisation of the truth table produces its minimal representation, where '-' indicates species responses that can take either truth value (b). Each row of the minimised truth table corresponds to a set of logical implication statements (c). Statements are chained together to create a logical implication network, which summarises the deterministic relationships between species responses as predicted by the model (d). The nodes represent species responses, and the arrows between nodes represent the direction of the logical implication, for example 'spp 3 − → spp 5 + ' reads 'if species 3 responds negatively then species 5 responds positively'. The small '− spp 3' in each node indicates that species are responding to a negative press-perturbation of species 3

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complete description (e.g. Figure 2b). Boolean minimisation is commonly used in circuit design, therefore algorithms are readily available to find a solution that is minimal (e.g. McCluskey, 1956) or close-to-minimal (e.g. Brayton, Hachtel, McMullen, & Sangiovanni-Vincentelli, 1984). The second step is to use each row of the minimised table to derive a set of logical implication statements (e.g. Figure 2c), which are deterministic rules governing the relationships between species responses. The final step is to connect implications together to construct a logical implication network (e.g Figure 2d). It can be shown (Theuns, 1994) that if at least one statement from every row of the minimised table is included, then the network gives a complete description of all species-response relationships in the model.

See Supporting Information C for tutorials on implementing the Boolean approach, and Supporting Information D for guidance on constructing and interpreting logical implication networks.

3 | MATERIALS AND METHODS

We analysed the predictions of the Macquarie Island model using two alternative methods: (a) Monte Carlo simulations (a standard QM approach expressing probabilistic PoI), which were used to test the effect of sampling method on the predictions, and (b) our new Boolean approach. We tested the ability of both approaches to handle interaction—strength uncertainty only: the structure of the ecological interaction network was fixed as shown in Figure 1a. We predicted species responses to rabbit control, using the sensitivity matrix approach, subject to three plausibility constraints: (a) the system is stable, (b) rabbits respond negatively to rabbit control, and (c) tall tussock vegetation responds positively to rabbit control.

3.1 | Monte Carlo simulations

The sampling method from Raymond et al. (2011) was used as a baseline for comparisons, and we evaluated the robustness of its predictions to three variations in the sampling method (Table 1). Tests 1 and 2 represent alternative but arguably equally defensible ways to sample interaction-strength values: test 1 is a reparameterisation of the method used in Raymond et al. (2011) and test 2 incorporates additional background information. Test 3 explores the effect of the sampling method on the stability constraint. Summary results were calculated from 500 plausible community matrices for tests 1 and 2, and 100 matrices per subsample for test 3.

3.1.1 | Test 1: Sampling in the Lotka–Volterra parameter space

Raymond et al. (2011) performed the Monte Carlo sampling on the elements of the community matrix. However, the community matrix represents the linearisation of some population dynamic model around its steady state, and is therefore a description of the parameter space in terms of derivative properties (Supporting Information B). When the parameter spaces have a physical meaning, a philosophical argument can be made that PoI should be applied to the primary description (Mikkelson, 2004). This raises Question 1: *do the species-response predictions change when the parameter space sampled is that of the primary description?*

To test this, we assumed that the community matrix is derived from the Lotka–Volterra model, and we performed the Monte Carlo simulations on the values of this model's parameters. The population dynamics are

$$\frac{dn_i}{dt} = n_i\left(r_i + \sum_{j=1}^{n} b_{ij}n_j\right), \tag{1}$$

where n_i is the population density of species i, r_i is the intrinsic growth rate of species i, and b_{ij} is the effect that species j has on species i. The parameter values sampled were r_i and b_{ij}. The time-scale t in Equation 1 can be chosen such that all $|r_i| \le 1$ and $|b_{ij}| \le 1$. Sampling on these parameters can produce a system with negative population density at steady state i.e. $n^* < 0$. Therefore we added the plausibility constraint $n^* > 0\ \forall i$.

TABLE 1 The different Monte Carlo sampling methods that were applied to the Macquarie Island case study to test the robustness of predictions. All tests included the following plausibility constraints: A is stable, $-S_{\text{rabbits,rabbits}} < 0$, $-S_{\text{tussock,rabbits}} > 0$. The parameters a_{ij} are non-zero elements of the community matrix, and b_{ij} and r_i are parameters from the equivalent Lotka–Volterra model

Name of test	Description of parameter space sampling method
Raymond baseline	Baseline method from Raymond et al. (2011) • $a_{ii} \sim \mathcal{U}(-1,-0.25)\ \forall i$ • $\lvert a_{ij}\rvert \sim \mathcal{U}(0.01,1)\ \forall i \neq j$
Test 1	Based on recommendation of Mikkelson (2004), which applies the Principle of Indifference to the Lotka–Volterra parameter space • $b_{ii} \sim \mathcal{U}(-1,0)\ \forall i$ • $\lvert b_{ij}\rvert \sim \mathcal{U}(0,1)\ \forall i \neq j$ • $r_i \sim \mathcal{U}(-1,1)$ • Additional plausibility constraint: $n_i^* > 0\ \forall i$
Test 2	Invokes background information that the interaction-strength distribution is skewed (e.g Neutel et al., 2002). • $a_{ii} \sim \mathcal{U}(-1,0)\ \forall i$ • If species i is a prey and species j is a predator: • $a_{ij} \sim \mathcal{U}(-1,0)$, and • $a_{ji} \sim \text{Beta}(\alpha,1)$ such that $\mathbb{E}[a_{ji}] = 10^{-2}a_{ij}$, otherwise: $\lvert a_{ij}\rvert \sim \mathcal{U}(0,1)$
Test 3	Explores the effect of 'strengthening' the stability constraint • If i is basal: $a_{ii} \sim \mathcal{U}(-1,0)$ otherwise: $a_{ii} \sim -\text{Beta}(1,\beta)$, and explore effect of $\mathbb{E}[a_{ii}]$ • $\lvert a_{ij}\rvert \sim \mathcal{U}(0,1)\ \forall i \neq j$

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3.1.2 | Test 2: Sampling from a skewed interaction-strength distribution

Raymond et al. (2011) sampled interaction strengths from a uniform distribution. However, detailed studies of real food webs show that interaction-strength distributions are not random and uniform, but rather have skewed distributions with many weak interactions (Wootton & Emmerson, 2005). This is often attributed to the size difference between predators and prey and feeding inefficiencies (Emmerson & Raffaelli, 2004; Loeuille & Loreau, 2005; Williams & Martinez, 2000). Background information like this is typically incorporated into the sampling distribution (Kass & Wasserman, 1996), and previous QM studies have also sampled interaction strengths from a skewed distribution (e.g. Hosack et al., 2008). This raises Question 2: *do the species-response predictions change when interaction strengths are sampled from a skewed distribution?*

To test this, we sampled the predator-to-prey interaction-strength magnitudes from a standard uniform distribution. Then, prey-to-predator magnitudes were sampled from a beta distribution, with a mean scaled to the predator-to-prey magnitude to reflect the heuristic that the former is approximately two orders of magnitude larger than the latter (Neutel, Heesterbeek, & de Ruiter, 2002).

3.1.3 | Test 3: Exploring the effect of stability strength

Raymond et al. (2011) used the stability of the community matrix as a plausibility constraint. This is justified by studies showing that webs have an idiosyncratic structure peculiarly suited to promoting stability (De Ruiter, Neutel, & Moore, 1995; Yodzis, 1981). However, the 'strength' of the stability constraint can be manipulated by changing the magnitude of the diagonal elements of the community matrix (representing species' self-limitation) relative to the off-diagonal elements. Negative diagonal elements stabilise the system; an unstable system can be converted to a stable one by arbitrarily increasing the magnitude of negative diagonal elements (c.f. Neutel et al., 2002; James et al., 2015). Raymond et al. (2011) (and also Harvey et al., 2016) sampled the off-diagonal elements from $|a_{ij}| \sim \mathcal{U}(0.01,1)$, but the diagonal elements were sampled from higher magnitudes $a_{ii} \sim \mathcal{U}(-1,-0.25)$. This raises Question 3: *do the species-response predictions change if the magnitude of the diagonal elements, and hence the 'strength' of the stability constraint, is changed?*

To test this, we sampled the consumer species' diagonal elements from a beta distribution, and explored the effect of increasing the strength of the stability constraint by decreasing that distribution's expected value. We chose the consumer species because they generally have low self-regulation compared to basal species and compared to other interaction strengths generally (by roughly one order of magnitude, see Christianou & Kokkoris, 2008; Emmerson & Raffaelli, 2004).

3.2 | Boolean approach

To construct the truth table, we performed a parameter-value sweep of the same Macquarie Island model that was used in the Monte Carlo simulations above. We performed the parameter-value sweep on the magnitudes of the interaction strengths, by sampling $|a_{ij}| \sim \mathcal{U}(0,1)$, to create a large sample of possible community matrices. We tested each matrix against the three plausibility constraints, and if the matrix passed, we used the sensitivity matrix to predict species responses to rabbit control. Every species—response combination that was observed at least once was identified as possible, and we assumed that combinations that were never seen were impossible. We sampled a total of 1×10^5 community matrices, and no new species–response combinations were found after the first 763 samples. This suggests that we found every combination that was possible, but it does not guarantee it (see Discussion).

To minimise the truth table, we used the Espresso algorithm implemented in the PyEDA 0.28 package (Drake, 2013) in Python 3 (see Data accessibility). Every logical implication statement with a single antecedent was included in the logical implication network.

Note that the distribution we used to perform the parameter-value sweep does not represent a prior, and the fact that it is uniform is not an expression of PoI. The uniform distribution was chosen for convenience only. Provided that every possible species—response combination is found, then the method used to find them does not influence the Boolean analysis.

4 | RESULTS

Alternative Monte Carlo sampling methods yielded quite different predictions for the same species response (selected examples in Figure 3, other species in Supporting Information E). For example, the proportion of negative responses in penguins decreased from 81% when the elements of the community matrix were sampled uniformly (test 3 with $\mathbb{E}[a_{ii}] = 0.5$), to 70% in the reimplementation of Raymond et al. (2011) sampling (Raymond baseline), to 64% when the Lotka–Volterra coefficients were sampled instead (test 1). And while under the baseline parameterisation both skuas and Antarctic prions had ambiguous responses, under parameterisation via the skewed distribution (test 2), there was high evidential support for a negative response by skuas and a positive response by prions. When the stability constraint was 'strengthened' (test 3), the proportion of matrices rejected increased as expected (Figure E.13), and some species' responses went from having high to ambiguous support (e.g. redpolls starlings from 90% to 46%), while others went from having ambiguous to high support (e.g. skuas from 42% to 94%).

The Boolean approach produced a minimised truth table with 12 rows, from which the logical implication network in Figure 4 was derived. Herbfield, albatrosses, and burrow-nesting seabirds respond positively to rabbit control (Figure 4a), consistent with the 100%

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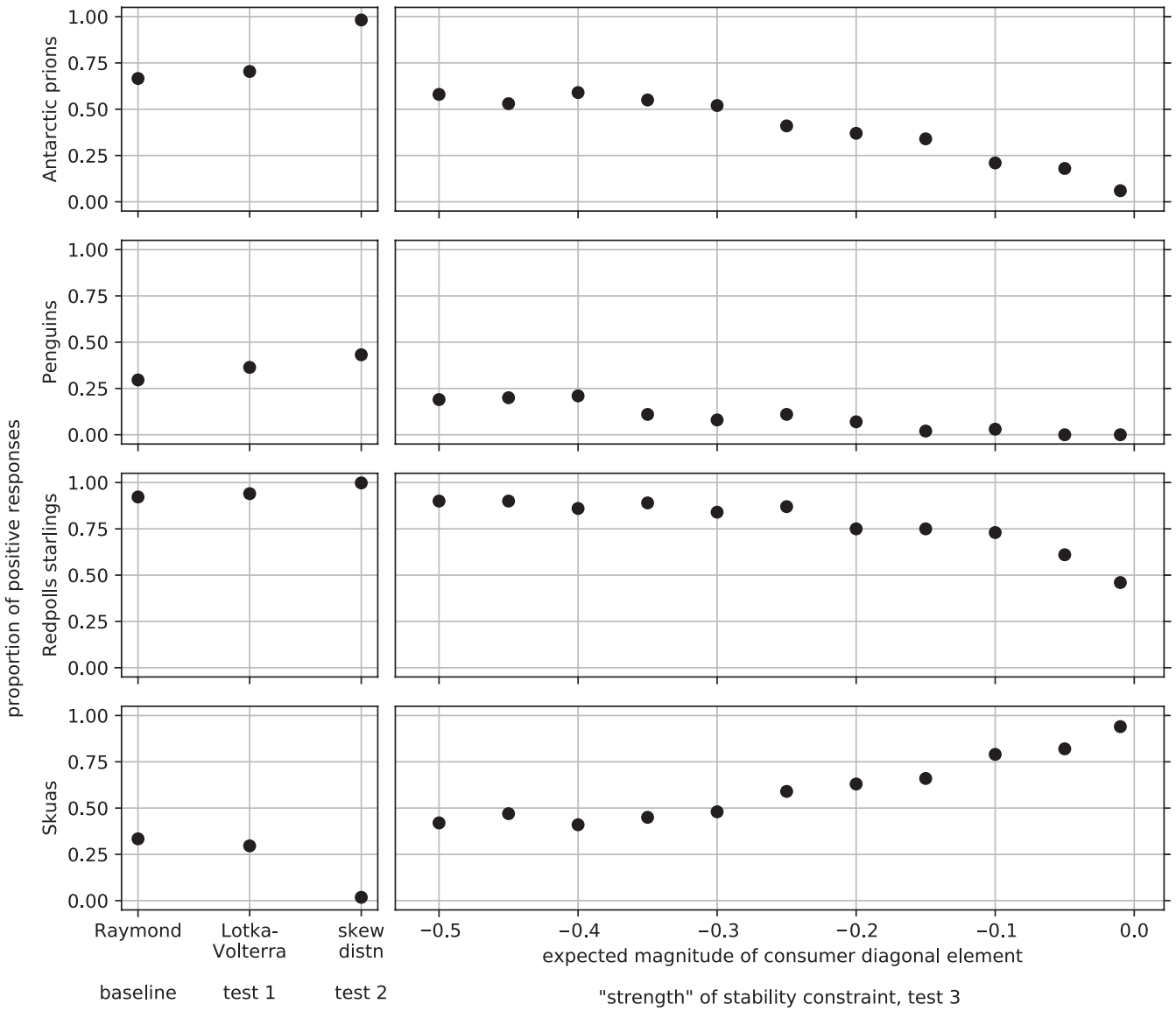

FIGURE 3 Selected results for Macquarie Island from the different Monte Carlo simulation methods in Table 1 (probabilistic QM). Predictions for some species differed between methods and were sensitive to the strength of the stability constraint. Additional results in Supporting Information E

probabilities for those responses obtained in Monte Carlo simulations (Supporting Information E). For some responses where Monte Carlo results were conflicting, the Boolean approach elucidated simple rules: penguin response is partially determined, such that if skuas respond positively then penguins respond negatively (Figure 4c); and the response of skuas is always opposite in sign to that of prions (Figure 4d).

Table 2 notes the computational time for both approaches.

5 | DISCUSSION

Qualitative modelling can be used to predict ecosystem behaviour even when species interaction strength data are lacking. However, philosophical and technical difficulties can arise if QM invokes the probabilistic form of the PoI. To illustrate this, we used a case study from Macquarie Island, with the goal of predicting species responses to pest control. We found that different but equally defensible ways of representing interaction—strength uncertainty can produce contradictory predictions. We drew from the philosophy literature to develop a non-probabilistic approach that eliminated the cause of these contradictions. We used Boolean analysis to reveal simple rules governing the model predictions that were not apparent from probabilistic QM.

Under a probabilistic PoI, alternative ways of specifying a QM problem can give rise to contradictory predictions, with differences that can be large enough to impact conservation decision-making. We demonstrated this in the Macquarie Island case study, where some species' responses to pest control were predicted with strong support by one Monte Carlo sampling method but

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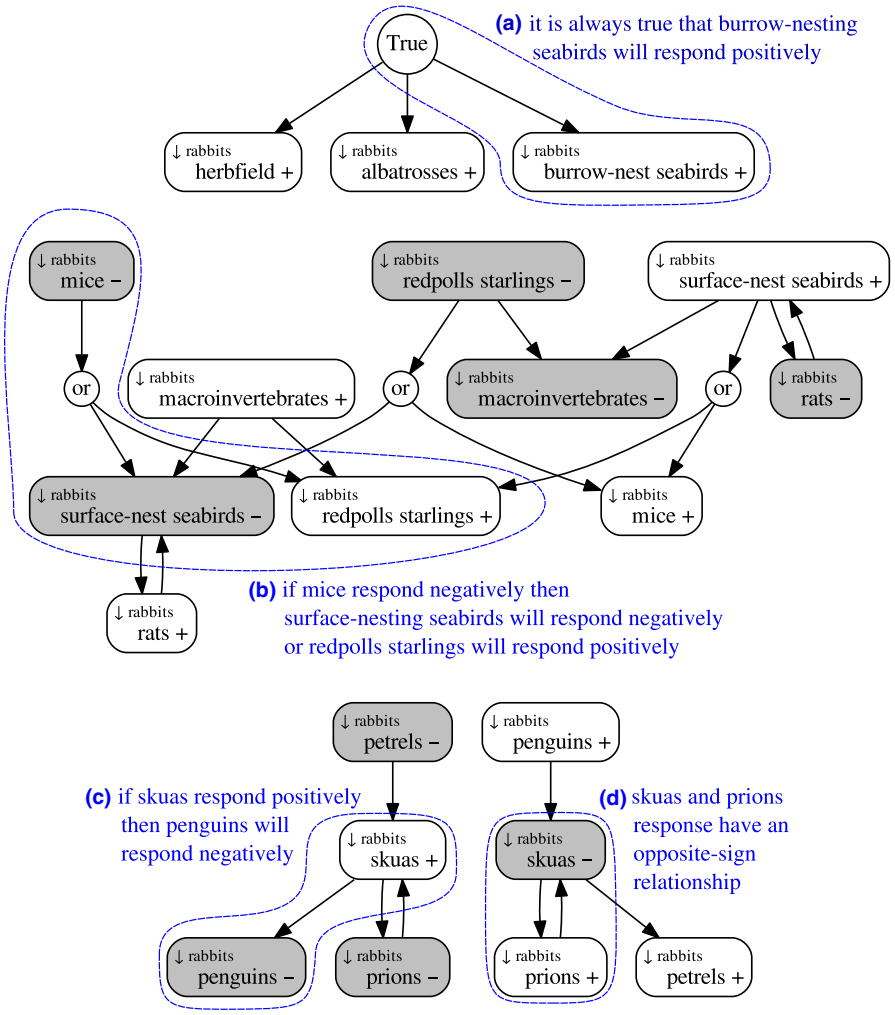

FIGURE 4 Logical implication network for Macquarie Island from the Boolean approach (non-probabilistic PoI). Species whose responses were ambiguous according to the Monte Carlo simulations or differed between sampling methods (Figure 3) are governed by simple deterministic rules. See Supporting Information D for more guidance interpreting network

were ambiguous according to another (Figure 3). We also found that the 'strength' of the stability constraint, which is necessary to capture the idiosyncratic interaction-strength patterns in real webs (e.g. De Ruiter et al., 1995), changed predictions (c.f. Section 2 of Kirk, Rolando, MacLean, & Stumpf, 2015). The point of creating alternative sampling methods was not to criticise the method that Raymond et al. (2011) chose; we do not believe that their choice was unreasonable. Rather, the point is that—under high uncertainty—many different yet arguably equally defensible sampling methods can be devised, and they can lead to very different predictions. Previous authors have argued that modelling uncertainty using a distribution is paradoxical because more information is needed to describe a distribution than an expected value (Rochet & Rice, 2009). Here, we have shown how trying to specify some 'neutral' distribution, expressing ignorance of the parameter values, can lead to paradoxical results (c.f. Kraak, Kelly, Codling, & Rogan, 2010).

It is not surprising that different sampling methods will produce different predictions, and under a Bayesian framework, these differences have a natural prior–posterior interpretation. This interpretation is useful for certain research questions. For example, May (1972) found that complex food web models with randomly sampled interaction strengths have a low probability of being stable, which prompted the wide-ranging complexity–stability debate, where the probabilistic approach remains a key tool (Landi, Minoarivelo, Brännström, Hui, & Dieckmann, 2018). However, in the context of conservation decision-making, the goal is to predict the behaviour of a particular system and in a way that accounts for experts' uncertainty. When experts' uncertainty is very high, then the best choice of parameter space and sampling distribution can also be uncertain; this aspect of uncertainty is what our Boolean approach addresses (Example 3 in Box 1).

In contrast to probabilistic QM, the Boolean approach produces deterministic predictions that are always true in the model, regardless of the parameter values. By invoking PoI in a non-probabilistic manner, our method generates robust predictions that subsume the contradictory predictions produced by different probabilistic methods. In the case study, different Monte Carlo sampling methods gave different predictions for Antarctic prions and skuas, but they always predicted that the percentage of positive responses in Antarctic prions was 100% minus the percentage of positive responses in skuas. Our Boolean approach revealed why: the response of prions is always opposite in sign to that of skuas. As another example, the response of penguins was ambiguous and

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TABLE 2 A comparison ranking various aspects of four QM methods' performance for modelling with interaction—strength uncertainty (impossible < worst < poor < good < best), and computation times at key algorithmic bottlenecks from the Macquarie Island case study. Computation times for Lenovo ThinkPad X1 Carbon with Intel i7 CPU (2 GHz) and 8 GB RAM

Aspect	Loop analysis (e.g. Lane & Levins, 1977)	Weighted-predictions matrix (e.g. Dambacher et al., 2011)	Monte Carlo simulations (e.g. Raymond et al., 2011)	Boolean approach (this paper)
Large models and computational time constraints?	*Worst to impossible* Requires symbolic calculation of matrix adjugate plus direct analysis of result	*Poor to* [a]*impossible* Matrix permanent cannot be calculated for large problems	*Best* Approximately 1000 plausible-matrix samples are sufficient to estimate expected value	*Poor to* [a]*impossible* Cannot perform Boolean minimisation for large problems. If using large parameter-value sweep, then cannot know when all response combinations found, and problem is worse for larger models
Use past responses as plausibility constraints?	*Good* Can incorporate constraints into direct analysis	*Impossible* Cannot use	*Best* Can easily filter matrices sampled against constraints	*Best* Can easily filter matrices accepted against constraints
If high uncertainty, no data available?	*Best* No PoI problems. Usefulness of predictions depends upon context and analytic tractability	*Poor* PoI problems for indeterminate responses. Single-species predictions only	*Worst* PoI problems (see text)	*Best* No PoI problems. Multispecies relationships predicted
If low uncertainty, data available?	*Poor* Relative constraints (e.g. $a_{ij} > a_{kl}$) easily incorporated, but only prior distribution's support can be used	*Impossible* Neither relative constraints nor prior distribution can be used	*Best* Relative constraints easily incorporated, and natural Bayesian (prior-posterior) interpretation	*Poor* Relative constraints easily incorporated, but only prior distribution's support can be used
Key computation times for Macquarie Island	NA	[a]T matrix: 14 s	1×10^3 matrices: 0.6 s	[a]1×10^5 matrices: 1 min [a]Minimisation: 1 s

[a]Computation time will grow exponentially with the size of the model (i.e. both matrix permanent (Valiant, 1979) and Boolean minimisation (Buchfuhrer & Umans, 2011) are NP hard problems).

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depended upon the parameter space sampled. The Boolean analysis revealed more specifically that penguins' response is partly determined: if skuas respond positively then penguins will respond negatively.

The Boolean approach has pragmatic and philosophical advantages over probabilistic QM. One pragmatic advantage is that a deterministic prediction may be easier to interpret, particularly when the probabilistic prediction is ambiguous. Simple relationships between species responses may also be used to identify indicator species for a post-intervention monitoring, or as candidates for additional intervention. For example, the determining effect of the rat response upon surface-nesting seabirds makes rats a good candidate as an indicator species. Alternatively, the positive effect of decreasing rats may inspire additional Boolean analysis, to investigate the outcome of a rat control programme (Supporting Information C.3). The Boolean approach can also identify key species that determine the overall system outcomes. In the theoretical example (Figure 2), species responses that were maximally ambiguous according to a weighted-predictions matrix analysis (Supporting Information A) were shown by the Boolean approach to constitute two separate regimes of system behaviour. In the Macquarie Island case study, clusters of species that are linked in the implication network indicate those that respond together. The Boolean approach can also produce results that are difficult to interpret, when there are few rules or all rules are long (Supporting Information D); however, this result useful in its own right because it indicates that the system is intrinsically difficult to predict.

The philosophical advantage of the Boolean approach is that it addresses the root cause of contradictory predictions in probabilistic QM. Contradictory predictions occur for the same reason as in philosophical thought-experiments: there is insufficient background information about the system. The missing background information is not merely ignorance of the parameter values, which PoI can work well to address, but rather a more fundamental ignorance about the nature of the system itself. PoI works well when the mechanism of randomisation is well understood, for example the toss of a die implies randomisation over the parameter space of the six faces. However, the mechanism that determines interaction-strength values involves an interplay between physical, biological, ecological and evolutionary processes, and these processes are poorly understood. Probabilistic QM forces us to specify the parameter space and randomisation anyway—in order to proceed with the modelling exercise—whereas the Boolean approach provides a way to produce predictions without implicitly overstating our knowledge.

The idea of obtaining deterministic predictions from community matrix models is not new. Loop analysis and the weighted-predictions matrix can identify if a species' response is guaranteed to be positive or negative or if it is indeterminate (i.e. both signs possible). However, these methods typically consider each species' deterministic response independently from other species. When species are considered independently, even for modestly large networks, most responses will be indeterminate (this motivated the probabilistic interpretation (Dambacher, Li, & Rossignol, 2002)). In contrast, our Boolean approach considers the interdependencies between species responses, which uncovers new deeper levels of determinacy (e.g. Supporting Information A).

The Boolean approach can only tell us about the predictions of a model, and we have not explored model validation. For example, we did not consider variations on the interaction network structure (i.e. variations on Figure 1a), which is an area for future work. We also treated the model assumptions as given; the Boolean approach cannot tell us if the 'near-equilibrium' assumptions (Abrams, 2001; Justus, 2006) are sufficient for modelling pest control. However, the separation between Boolean analysis and the model means that it can be applied to any model or system where the response variables have a natural dichotomisation, including future improvements to pest control models, and also other kinds of models (e.g. social–ecological, Supporting Information C.4).

Finally, we highlight two caveats to the Boolean approach: (a) the assumption that the parameter-value sweep uncovered *all* possible response combinations for input to the Boolean analysis, and (b) the fact that Boolean minimisation is itself computationally expensive. Both caveats mean that, for sufficiently large models, a Boolean approach will be impossible (Table 2).

The first caveat arises because our Boolean analysis assumes that we sampled the parameter space sufficiently to obtain all possible response combinations. In our case study, we strongly suspect that we achieved this because the number of samples was two orders of magnitude larger than was needed to find the final new combination. However, when the parameter space is continuous, then the number of possible parameter values is infinite, therefore there is no guarantee that a naive parameter-sweep will find all possible combinations. It is possible, for certain kinds of problems, to identify the finite number of points at which the parameter values must be sampled to obtain all response combinations (e.g. Giordano, Samaniego, Franco, & Blanchini, 2016; Sontag, 2014). Perhaps these methods can be extended to more general cases in the future. In the meantime, however, because this problem gets worse the larger and more connected the interaction network, is it remains a limitation of our approach and may prevent its use for complex models.

Second, Boolean minimisation is an NP hard problem; roughly speaking, this means that the computation time required increases exponentially with the size of the problem. The practical consequence is that it is not possible to minimise the full truth table and obtain the logical implication network when the number of responses of interest is 'large'. To give an example of what can be considered 'large', minimising 16 species responses in a Christmas Island food web (unpublished data) was the approximate computational limit. Furthermore, in multispecies pest control, the effects of each pest's suppression on a species cannot be summed (Supporting Information F) but must instead be treated separately, and therefore, the problem size doubles with each additional pest suppressed. This limitation is intrinsic to Boolean minimisation.

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We are faced with an urgent biodiversity crisis, and predicting complex ecosystem behaviour to stem further species losses is difficult when data are lacking. The methods we have examined were a constructive response to this pressing need, and it was only by merit of their transparency and theoretically rigorous basis that our work was possible. We hope that highlighting their potential problems will not be discouraging, but instead inspire further technical development to meet the important conservation challenges ahead.

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AUTHORS' CONTRIBUTIONS

E.M.M. conceived the problem and acquired the funding; E.M.M. and R.A.C. contributed to the writing; N.P.K. conceived the Boolean approach, analysed all the models, and led the writing. All authors gave final approval for publication.

DATA ACCESSIBILITY

Data are included with code and tutorials, which are archived at https://doi.org/10.5281/zenodo.2574324.

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